

# Ahmed Hendawy

*Ph.D. Student in Reinforcement Learning*

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## About me

I am a fifth-year Ph.D. student in the LiteRL and IAS research groups at TU Darmstadt, Germany, supervised by Prof. Carlo D'Eramo and Prof. Jan Peters. My research focuses on reinforcement learning (RL), where I develop scalable, broadly applicable methods to improve agent learning, with a particular emphasis on multi-task RL and model composition and interaction. My work explores representation learning, mixture-of-experts, model merging, and the optimization structure of RL objectives to enhance generalization, robustness, and sample efficiency. I have also contributed to foundational deep RL algorithms that improve stability and learning efficiency. Currently, I investigate large-scale multi-task learning and the integration of Vision-Language-Action (VLA) models into RL systems, drawing on ideas from both multi-task RL and large language models (LLM) research.

## Education

- 2022–present **Ph.D. in Computer Science. Advisors: Carlo D'Eramo & Jan Peters**, *Technical University of Darmstadt*, Darmstadt - Germany, Expected graduation: Dec 2026.
- 2019–2021 **M.Sc. in Information Technology**, *University of Stuttgart*, Stuttgart - Germany.  
Specialization: *Computer Engineering*, GPA: 1.5 (German scale, 1.0 = best)
- 2021 **Master's Thesis**, *University of Stuttgart & Robert Bosch GmbH*, Stuttgart - Germany.  
Title: *Constraint-based Optimization Approach for Generalized Few-Shot Object Detection*, GPA: 1.0 / 1.0
- 2014–2019 **B.Sc. in Mechatronics Engineering**, *German University in Cairo*, Cairo - Egypt.  
GPA: 0.82 / 0.70 (Excellent with the Highest Honors, ranked 2nd in class).
- 2018 **Bachelor's Thesis**, *Technical University of Munich*, Munich - Germany.  
Title: *A Hybrid Approach for Constrained Deep Reinforcement Learning*, GPA: 1.0 / 1.0

## Technical Skills

- Programming Python, C++, Java
- Frameworks PyTorch, JAX, TensorFlow / Keras, ROS1 / ROS2.
- Simulators MuJoCo, Gazebo

## Work Experience

- 04/2022–present **Doctoral Researcher**, *Reinforcement Learning*, LiteRL - IAS, TU Darmstadt, Germany.  
My responsibilities include conducting research, publishing scientific papers, teaching, and supervising bachelor's and master's students.
- 11/2021–12/2021 **Graduate Student Researcher**, *Computer Vision*, Robert Bosch GmbH, Germany.  
I prepared my master's thesis work for publication as a research paper.
- 11/2020–04/2021 **Research Intern**, *Computer Vision*, Robert Bosch GmbH, Germany.  
I investigated methods to improve existing few-shot object detection algorithms and built a reliable codebase for my master's thesis.
- 04/2020–10/2020 **Graduate Student Researcher**, *Machine Learning*, Fraunhofer IPA, Germany.  
I proposed a deep learning approach for *Material Identification* by denoising Radar raw signals using a *Wavenet* network.
- 04/2020–07/2020 **Graduate Student Researcher**, *Software Engineering*, Institute of System Dynamics, University of Stuttgart, Germany.  
I developed a Python library to wrap the linear optimization algorithms written in Julia (PyLinearSolver).

- 09/2019– **Developer Advocate**, *Software Engineering*, Avelabs, Egypt.
- 11/2019 I developed ADAS/AI applications for autonomous driving in general and perception systems in particular, using the company's product "Yonohub," which is a cloud-based system for designing, sharing, and evaluating complex systems, such as Autonomous Vehicles, ADAS, and Robotics. In addition, I wrote blog posts on Medium to advocate for the product.
- 06/2019– **Developer Advocate Intern**, *Software Engineering*, Avelabs, Egypt.
- 09/2019 I developed a platform using Yonohub to benchmark multi-object tracking algorithms (e.g., AB3DMOT) against the nuScenes dataset.

## Publications & Preprints

### Conference Papers

- 2026 **Ahmed Hendawy**, Henrik Metternich, Théo Vincent, Mahdi Kallel, Jan Peters, and Carlo D'Eramo. Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning. *International Conference on Learning Representations (ICLR)*, 2026.
- 2024 **Ahmed Hendawy**, Jan Peters, and Carlo D'Eramo. Multi-Task Reinforcement Learning with Mixture of Orthogonal Experts. *International Conference on Learning Representations (ICLR)*, 2024.
- 2023 Karim Guirguis, Mohamed Abdelsamad, George Eskandar, **Ahmed Hendawy**, Matthias Kayser, Bin Yang, and Juergen Beyerer. Towards Discriminative and Transferable One-Stage Few-Shot Object Detectors. *Winter Conference on Applications of Computer Vision (WACV)*, 2023.

### Journal Papers

- 2025 Joe Watson, Chen Song, Oliver Weeger, Theo Gruner, An T. Le, Kay Hansel, **Ahmed Hendawy** et al. Machine Learning with Physics Knowledge for Prediction: A Survey. *Transactions on Machine Learning Research (TMLR)*, 2025.

### Under Review

- 2026 **Ahmed Hendawy**, Henrik Metternich, Jan Peters, and Carlo D'Eramo. Analyzing Value-Based Reinforcement Learning through the Lens of Mode Connectivity. *Under Review*.
- 2026 Mahdi Kallel, Johannes Tölle, **Ahmed Hendawy**, and Carlo D'Eramo. Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement. *Under Review*.

### Workshop Papers

- 2025 **Ahmed Hendawy**, Gabriele Tiboni, Jan Peters, and Carlo D'Eramo. It is All Connected: Multi-Task Reinforcement Learning via Mode Connectivity. *Eighteenth European Workshop on Reinforcement Learning*.
- 2023 Marcel Mittenbuehler, **Ahmed Hendawy**, Carlo D'Eramo, and Georgia Chalvatzaki. Parameter-efficient Tuning of Pretrained Visual-Language Models in Multitask Robot Learning. *In CoRL Workshop on Learning Effective Abstractions for Planning (LEAP)*, 2023.
- 2023 Henrik Metternich, **Ahmed Hendawy**, Pascal Klink, Jan Peters, and Carlo D'Eramo. Using Proto-Value Functions for Curriculum Generation in Goal-Conditioned RL. *In NeurIPS Workshop on Goal-Conditioned Reinforcement Learning*, 2023.
- 2022 Karim Guirguis, **Ahmed Hendawy**, George Eskandar, Mohamed Abdelsamad, Matthias Kayser, and Jürgen Beyerer. CFA: Constraint-based finetuning approach for generalized few-shot object detection. *In CVPR Workshop on Learning with Limited Labelled Data for Image and Video Understanding (L3D-IVU)*, 2022.

## Honors & Awards

- 2022 **Best Paper Runner-up Award at the CVPR 2022 Workshop on Learning with Limited Labeled Data for Image and Video Understanding**.  
Paper title: CFA: Constraint-based Finetuning Approach for Generalized Few-Shot Object Detection
- 2022 **Sony Graduate Award for Best Master's Thesis**.  
Thesis title: Constraint-based Optimization Approach for Generalized Few-Shot Object Detection
- 2020-2021 **The Deutschlandstipendium (DStip) Scholarship at the University of Stuttgart**.  
Half of the funding of the D-Stips comes from the Federal Ministry of Education and Research (BMBF), the other half is from private donors, STIHL Germany.

- 2014-2019 **Full Scholarship for Bachelor of Science in Mechatronics Engineering at the German University in Cairo.**  
I ranked second in the Suez governorate during high school, which was encouraging for the German University in Cairo to offer me a full scholarship for a Bachelor degree in Engineering at the university.
- 2019 **1<sup>st</sup> Place of the Technical Innovation Award in Shell Eco-Marathon Asia 2019.**  
I led an R&D team that developed a system for vehicle drivers with physical limitations (such as missing arms or legs) using a drive-by-wire system that is controlled via brainwave and facial sensors.
- 2018 **Scholarship for Bachelor Thesis at The Technical University in Munich.**  
I was awarded a scholarship by the German University in Cairo to conduct my Bachelor Thesis at the Technical University in Munich due to my academic performance (Ranked 2<sup>nd</sup> among 251 students in the class of Mechatronics Engineering).
- 2018 **2<sup>nd</sup> Place of the safety award in Shell Eco-Marathon Asia 2018.**  
Using heart rate sensors, face recognition, rain sensors, and other critical sensors inside the car to develop a safety system for the driver and the vehicle.
- 2018 **1<sup>st</sup> place at the iCompete Competition of iHub at Ain Shams University.**  
An AGV Robot, called Solar Nanny, was designed, manufactured, and controlled to dry clean the surface of solar panels in industrial fields.

## Invited Talks

- 2025 **Multi-Task Reinforcement Learning via Mixture of Experts**, *Amii Presents: AI Seminar*.  
University of Alberta, Edmonton, Canada
- 2024 **Multi-Task Reinforcement Learning with Mixture of Orthogonal Experts**, *Mate Podcast*.
- 2024 **Introduction to Reinforcement Learning**, *To Data and Beyond Podcast*, In Arabic.

## Student Supervision

- 2026 **Henrik Metternich**, *Value-Based Reinforcement Learning via Mode Connectivity*, Master's Thesis.
- 2024 **Marcel Mittenbühler**, *Parameter-Efficient Tuning of Pretrained Visual-Language Models in Multitask Robot Learning*, Master's Thesis.
- 2024 **Lorenzo Tozzi**, *Dynamic Tiles for Deep Reinforcement Learning*, Master's Thesis.

## Teaching Experience

- 2025 **Teaching Assistant**, *Robot Learning Integrated Project*, Winter Semester, TU-Darmstadt, Germany.
- 2023 **Head of Teaching Assistants**, *Reinforcement Learning: From Fundamentals to the Deep Approaches*, Summer Semester, TU-Darmstadt, Germany.
- 2022 **Teaching Assistant**, *Reinforcement Learning: From Fundamentals to the Deep Approaches*, Summer Semester, TU-Darmstadt, Germany.

## Organization & Reviewing Activities

### Workshop Organization

- 07/2026 **Workshop on Reinforcement Learning for Vision-Language-Action Models (RL4VLA).**  
Robotics: Science and Systems (RSS) in Sydney, Australia
- 08/2025 **Workshop on Inductive Biases in Reinforcement Learning (IBRL).**  
Reinforcement Learning Conference (RLC) in Edmonton, Canada
- 07/2024 **International Workshop of Intelligent Autonomous Learning Systems (IWIALS).**  
Kleinwalsertal, Austria
- 12/2023 **1st Symposium On Lifelong Explainable Robot Learning (SYMLER).**  
Nürnberg, Germany

### Reviewing

- Conferences International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), Conference on Neural Information Processing Systems (NeurIPS), Reinforcement Learning Conference (RLC), Conference on Robot Learning (CoRL), Artificial Intelligence and Statistics (AISTATS) (**Top Reviewer**)

Workshops European Workshop on Reinforcement Learning (EWRL)

## Open-Source Software

**MINTO**, *Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning*.

GitHub: <https://github.com/AhmedMagdyHendawy/MINTO>

**MOORE**, *Multi-Task Reinforcement Learning with Mixture of Orthogonal Experts*.

GitHub: <https://github.com/AhmedMagdyHendawy/MOORE>

## Extracurricular Activities

2019 **R&D Team Lead**, *Shell Eco-Marathon GUC Team (GUC Innovators)*.

2018 **Project Manager of Dashboard and Accessories**, *Shell Eco-Marathon GUC Team (GUC Innovators)*.

2017 **Motor Control Team Member**, *Shell Eco-Marathon GUC Team (GUC Innovators)*.

## Languages

Arabic Native

English Professional proficiency

German Basic (A1)